ARTIFICAL INTELLIGENCE

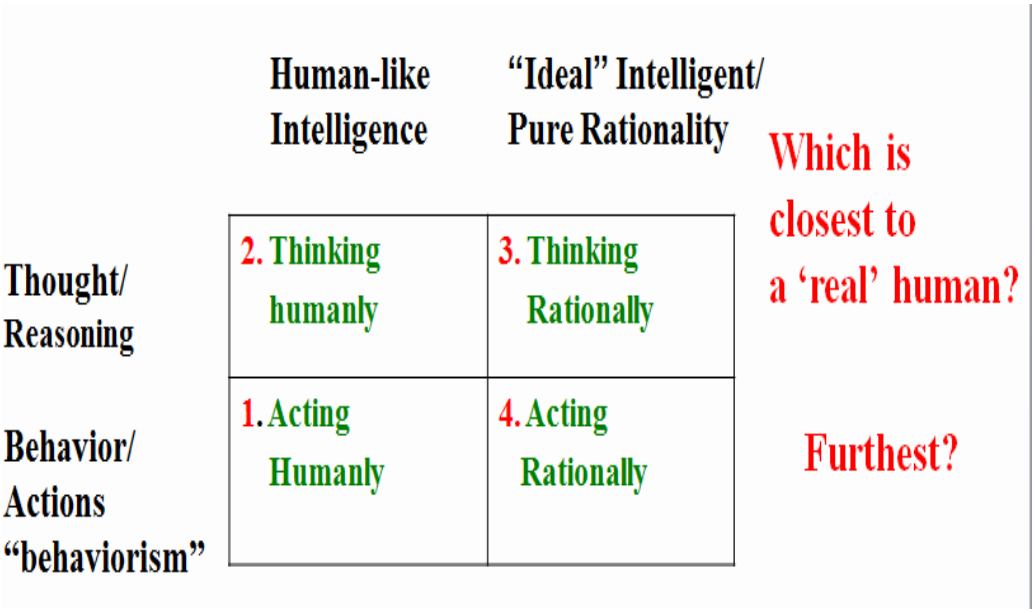
Definition of AI

"It is a branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions."

(without being explicitly pre-programmed, the system has to work with its own intelligence (man-made thinking power).

- Some definitions measure success in terms of human performance, whereas some measure against ideal concept of intelligence, that is, rationality. A system is rational if it does right things.
- This gives four possible goals to peruse AI. So definitions are organized into four categories with **two dimensions:**
  1) Thinking versus Acting
  2) Human versus Rational

1. **Systems that think like humans:** The exciting new effort to make computers as thinking machines with brain.
2. **Systems that think rationally:** The study of the computations that make it possible to perceive, reason, and act correctly.
3. **Systems that act like humans:** The study of how to make computers do things at which, at the moment, people are better.
4. **Systems that act rationally**: The branch of computer science that is concerned with the automation of intelligent behaviour.

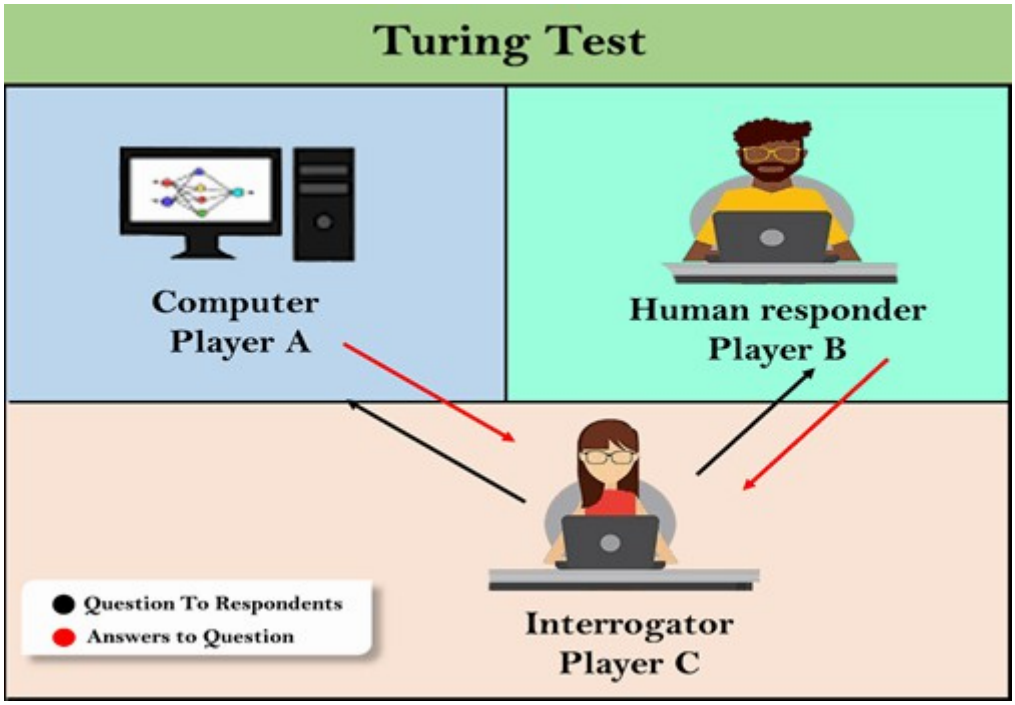


Acting humanly: The Turing Test approach

- **The Turing Test,** proposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence.
- Turing defined intelligent behaviour as the ability to achieve human- level performance in all cognitive tasks, sufficient to fool an interrogator.

Usually the term "cognitive task(consicousnss)" is used in the context of "cognitive task analysis." This is the process of analyzing or testing what level of thought process has reached or is using to solve a specific problem. You might look at a toddler, for example, who touches a hot iron and gets burned. Examples selecting the "odd man out" among three or more alternatives, identifying whether a single presented number (or letter) was or was not part of a previously presented set,

- The test he proposed is that the computer should be interrogated by a human via a teletype, and passes the test if the interrogator cannot differentiate responses between a computer or a human at the other end.



The computer would need to possess the following capabilities:

- **natural language processing** helps computers to understand the way that humans write and speak and enables it to communicate successfully with an intelligent system using a natural language like English (or some other human language) and making the machine to understand, analyse, manipulate, and interpret human's languages.
  - **knowledge representation** to store information provided before or during the interrogation; It is responsible for representing information about the real world so that a computer can **understand and can utilize this knowledge to solve the complex real world problems** such as diagnosing a medical condition or communicating with humans in natural language. Knowledge representation is not just storing data into some database, but it also enables an intelligent machine to **learn from that knowledge and experiences** so that it can behave intelligently like a human.

- **automated reasoning**, Reasoning plays a great role in the process of artificial Intelligence. Thus Reasoning can be defined as the logical process of **drawing conclusions, making predictions** or constructing approaches towards a particular thought with the **help of existing knowledge**. In artificial intelligence, reasoning is very important because to understand the human brain, how the brain thinks; how it draws conclusions towards particular things for all these sorts of works we need the help of reasoning.
- **machine learning** is the study of computer algorithms that can improve automatically through experience and by the use of existing data and to adapt to new circumstances and to detect and derive patterns.

Turing's test deliberately avoided direct physical interaction between the interrogator and the computer, because physical simulation of a person is unnecessary for intelligence.

Thinking humanly: The cognitive modelling approach

- cognitive computing is typically used to describe **AI systems that aim to simulate human thought**.
- Thinking humanly means trying to understand and model how the human mind works.

• There are (at least) two possible routes that humans use to find the answer to a question:

– We reason about it to find the answer. This is called "**introspection".**

– We conduct **psychological experiments** to find the answer, drawing upon scientific techniques to conduct controlled experiments and measure change. To say that a given program thinks like a human, we must have some **way of determining how humans think**. We need to **get inside** the actual workings of human minds to see how it works and then comparing our computer programs to this. This is what cognitive science attempts to do.

- The field of Cognitive Science focuses on modeling how people think, **computer models + experimental techniques from psychology**

Once we have a sufficiently precise theory of the mind, it becomes possible to express the theory as a computer program. If the program's input/output and timing behaviour matches human behaviour, that is evidence that some of the program's mechanisms may also be operating in humans.

Thinking rationally: The laws of thought approach

- Trying to understand **how we actually think** is one route to AI.
- But another approach is to model how we should think.
- The "thinking rationally" approach to AI uses symbolic logic (Symbolic logic is an expression of logic by using symbols in the place of natural language using propositions, truth tables,) to **capture the laws of rational thought as symbols that can be manipulated.**

Symbolic logic example:

Propositions: If all mammals feed their babies milk from the mother (A). If all cats feed their babies mother's milk (B). All cats are mammals(C).

The Λ means “and,” and the symbol means “implies.”

Conclusion: A Λ B C

Explanation: Proposition A and proposition B lead to the conclusion, C. If all mammals feed their babies milk from the mother and all cats feed their babies mother’s milk, it implies all cats are mammals.

- Reasoning involves manipulating the symbols according to well-defined rules, kind of like algebra.
- The Greek philosopher Aristotle was one of the first to attempt to codify "right thinking," that is, irrefutable reasoning processes. His famous syllogisms provided patterns for argument structures that always gave correct conclusions given correct premises. For example, "Socrates is a man; all men are mortal; therefore Socrates is mortal." These laws of thought were supposed to govern the operation of the mind, and initiated the field of logic.

- There are two main obstacles to this approach.
  - First, it is difficult to **make informal knowledge and state it in the formal terms required by logical notation,** particularly when there is **uncertainty** in the knowledge.
  - Second, there is a big difference between being able to solve a problem "**in principle" and doing so in practice.**

Acting rationally: The rational agent approach

- Acting rationally means acting so as to achieve one's goals, given one's beliefs.
- An agent is just something that perceives and acts.
- In this approach, AI is viewed as the study and construction of rational agents.
- playing a game will act rationally if it tries to win the game
- In the ‘laws of thought" approach to AI, the whole emphasis was on correct inferences. This is often part of being a rational agent because **one way to act rationally is to reason logically and then act on ones conclusion**s.
- On the other hand, correct inference is not all of rationality, because there are often situations where there is **no provably correct thing to do**, yet something must still be done. There are also ways of acting rationally that cannot be reasonably said to involve inference. For example, pulling one's hand off of a hot stove is a reflex action that is more successful than a slower action taken after careful deliberation.
- The study of AI as rational agent design has an advantage that it is more general than the logical approach because correct inference is only a useful mechanism for achieving rationality, not a necessary one.

	Humanly	Rationally
Thinking	Thinking humanly — cognitive modeling. Systems should solve problems the same way humans do.	Thinking rationally — the use of logic. Need to worry about modeling uncertainty and dealing with complexity.
Acting	Acting humanly — the Turing Test approach.	Acting rationally — the study of rational agents: agents that maximize the expected value of their performance measure given what they currently know.

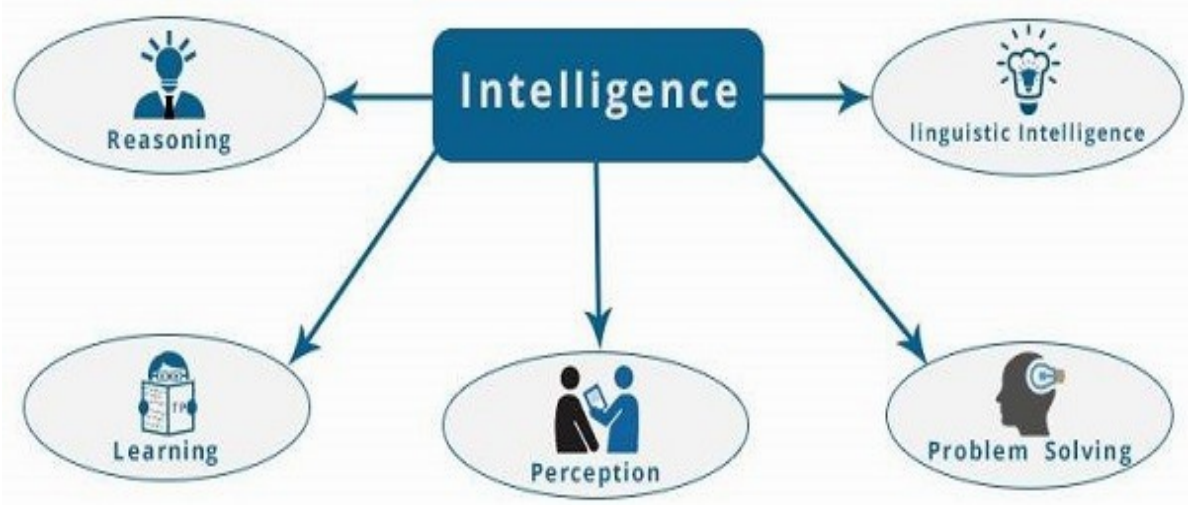
Components of AI

Goal of AI

Replicate/mimic human intelligence, Solve knowledge intense tasks with more accuracy and concise, precision results, is a connection of Perceptions and actions.

Driving car in traffic using sensors and image processing, Playing chess, doing surgical operations like human.

Can exhibit intelligence behaviour, lean new things by itself, demonstrate and explain, advise new things itself to user (why it has taken those decisions logical reasoning behind the decision taken)



Artificial Intelligence exists when a machine can have human based skills such as below

- **Perception**
- **Learning**
- **Linguistic intelligence**
- **Problem solving**
- **Reasoning**

Perception- It is the process of acquiring, interpreting, selecting, and organizing sensory information. Perception presumes sensing. In humans, perception is aided by sensory organs. In the domain of AI, perception mechanism puts the data acquired by the sensors together in a meaningful manner to take some decision.

Learning- Learning is the process of gaining knowledge by studying, understanding, practicing, being taught, or experiencing some thing. Learning enhances the awareness of any topic.

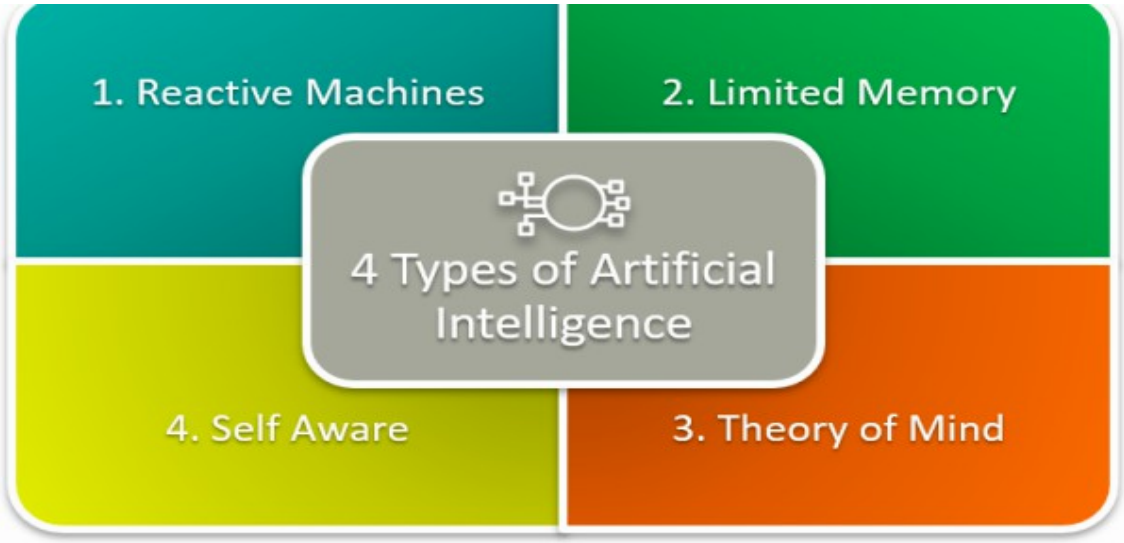
Linguistic intelligence- Linguistics- The ability to speak, recognize, and use mechanisms of phonology (speech sounds), syntax (grammar), and semantics (meaning).
Linguistic intelligence is one's ability to use, comprehend, speak, and written language. It is important in interpersonal communication.

Problem solving- Problem solving is the method during which **one perceives** and tries to make a **desired answer from a present** state of affairs by taking some path, that is blocked by known or unknown hurdles. It is the process of **selecting the best suitable alternative** out of multiple alternatives to reach the desired goal are available.

Reasoning- Reasoning is the set of processes that enables an intelligent system to help or to provide basis for judgement/actions, making decisions, and prediction.
Reasoning is of two types: Inductive Reasoning and Deductive Reasoning.

Inductive reasoning help in making generalization from specific facts and knowledge. Inductive reasoning is the bottom-up process. (goal--> steps followed to reach that goal)	It starts with a general statement uses available facts, information or knowledge and examines the possibilities to draw a specific, logical conclusion. (start with a step1 then use some steps to reach to goal)

Artificial Intelligence type-2: Based on functionality



1. **Reactive Machines**
   - Purely reactive machines are the most basic types of Artificial Intelligence.
   - Such AI systems **do not store memories or past experiences for future actions.**
   - These machines **considers only current scenarios and react on it as per possible best action**.They perceive the world and react to it.
   - Examples of Reactive AI is the famous **IBM Deep Blue system Chess program that beat the world champion**, Garry Kasparov, Deep Blue can make predictions about what moves might be next for it and its opponent. It ignores everything before the present moment and looks at the chessboard pieces as it stands right now and chooses from possible next moves and one more example is Google's AlphaGo

2. **Limited Memory**
   - Like the name suggests Limited Memory AI, can make informed and **improved decisions by studying the past data from its memor**y. Such an AI has a short-lived or a **temporary memory t**hat can be used to store past experiences and hence evaluate future actions, but they cannot add it to a library of their experiences.
   - Self-driving cars are one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road and uses sensors to identify civilians crossing the road, steep roads, traffic signals and so on to make better driving decisions. This helps to prevent any future accidents.

3. **Theory of Mind**
   - This category of machines is speculated to play a major role in psychology.
   - This kind of AI requires a thorough understanding that the people and things within an environment can alter feelings and behaviors. Theory of Mind AI should **understand people's emotions, sentiments, and thoughts and be able to interact socially like humans.**
   - Even though many improvements are there in this field, this kind of AI is not fully complete yet.

- One real-world example of the theory of mind AI is Kismet. **Kismet is a robot head** can mimic human emotions and recognize them. Both abilities are key advancements in theory of mind AI, but Kismet can't follow gazes or convey attention to humans.
- **Sophia** from Hanson Robotics is another example where the theory of mind AI was implemented. Cameras present in Sophia's eyes, combined with computer algorithms, allow her to see. She can sustain eye contact, recognize individuals, and follow faces, have emotions and feelings.

4. **Self-Awareness**
   - Self-Awareness AI does not exist in reality still and it is a hypothetical concept.
   - Such systems **understand their internal traits, states, and conditions and perceive human emotions. These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.** This type of AI will not only be able to understand and evoke emotions in those it interacts with, but also have emotions, needs, and beliefs of its own.
   - Self-awareness AI is the future of Artificial Intelligence. These machines will be smarter than human mind.

History of Artificial Intelligence

Artificial Intelligence is not a new word and not a new technology for researchers. This technology is much older than you would imagine. Even there are the myths of Mechanical men in Ancient Greek and Egyptian Myths. Following are some milestones in the history of AI which defines the journey from the AI generation to till date development.

Gestation of Artificial Intelligence (1943-1955)

- **Year 1943:** The first work which is now recognized as AI was done by Warren McCulloch and Walter pits in 1943. They proposed a model of **artificial neurons**.
- **Year 1949:** Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called **Hebbian learning**.
- **Year 1950:** The Alan Turing who was an English mathematician and pioneered Machine learning in 1950. Alan Turing publishes **"Computing Machinery and Intelligence"** in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior equivalent to human intelligence, called a **Turing test**.

The birth of Artificial Intelligence (1952-1956)

- **Year 1955:** An Allen Newell and Herbert A. Simon created the "first artificial intelligence program"Which was named as **"Logic Theorist"**. This program had proved 38 of 52 Mathematics theorems, and find new and more elegant proofs for some theorems.
- **Year 1956:** The word "Artificial Intelligence" first adopted by American Computer scientist John McCarthy at the Dartmouth Conference. For the first time, AI coined as an academic field.

At that time high-level computer languages such as FORTRAN, LISP, or COBOL were invented. And the enthusiasm for AI was very high at that time.

The golden years-Early enthusiasm (1952-1969)

- In AI's early years, limited computing power made even simple intelligent behavior seem remarkable. Many believed machines couldn't perform certain task. Newell and

Simon's early success was followed up with the General Problem Solver, or GPS. Unlike Logic Theorist, this program was designed from the start to imitate human problem-solving protocols.

- In 1958, John McCarthy created Lisp, introduced the Advice Taker for general AI, helped invent time sharing, and later advanced logic-based AI at Stanford, while Minsky focused on practical, anti-logic approaches.
- Minsky supervised a series of students who chose limited problems that appeared to require intelligence to solve. These limited domains became known as microworlds.

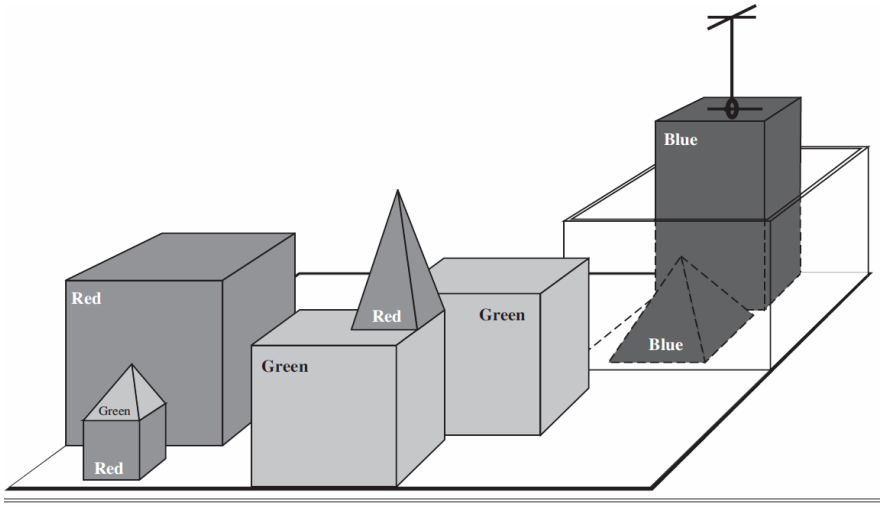


Figure 1.4 A scene from the blocks world. SHRDLU (Winograd, 1972) has just completed the command "Find a block which is taller than the one you are holding and put it in the box."

- **Year 1966:** The researchers emphasized developing algorithms which can solve mathematical problems. Joseph Weizenbaum created the first chatbot in 1966, which was named as ELIZA.
- **Year 1972:** The first intelligent humanoid robot was built in Japan which was named as WABOT-1.

A Dose Of Reality (1966–1973):

- In 1957, Herbert Simon boldly predicted that machines would soon match human intelligence in thinking, learning, and problem-solving.
- Early AI predictions, like Simon's, were overly optimistic due to initial successes on simple tasks. However, systems failed on real-world problems due to lack of background knowledge and computational complexity, as seen in flawed machine translation and slow theorem proving.
- In 1966, a report by an advisory committee found that "there has been no machine translation of general scientific text, and none is in immediate prospect."
- 1969: Minsky and Papert's Perceptron's book showed limitations of simple neural networks, stalling neural-net research. Backpropagation algorithms for multilayer networks were discovered (Bryson and Ho) but gained attention only decades later.
- 1973: The Lighthill Report criticized AI for combinatorial explosion; UK cut most AI research funding, limiting it to two universities.

Knowledge-based systems: The key to power? (1969–1979)

- 1975 – Minsky: Introduced frames for structured knowledge.
- 1976 – Rieger: Modeled human memory for language.

- 1977 – Schank & Abelson: Developed scripts for stereotypical events.
- 1978 – Wilensky: Studied goals and plans in language.
- 1981 – Schank & Riesbeck: Focused on memory-based language understanding.
- 1983 – Dyer: Simulated emotion and memory in story understanding.
- 1983 – Kolodner: Worked on case-based memory models.

AI becomes an industry (1980–present)

- The first successful expert system, R1, was used by Digital Equipment Corporation in 1982 to help configure computers, saving millions of dollars. By the late 1980s, many companies like DEC and DuPont were using expert systems, and nearly every big U.S. company had an AI team.
- The duration between years 1974 to 1980 was the first AI winter duration. AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.
- During AI winters, an interest of publicity on artificial intelligence was decreased.

The return of neural networks (1986–present)

- In the mid-1980s, researchers rediscovered the backpropagation algorithm (first created in 1969) and used it to train neural networks. This sparked interest in connectionist models—systems that learn like the brain by connecting many small units (neurons).
- These models were seen as rivals to symbolic AI, which used logic and rules. Some researchers argued that real thinking might not involve symbols at all. Today, most agree that both approaches—connectionist and symbolic—can work together.

AI adopts the scientific method (1987–present)

- AI research focuses on real-world applications, using rigorous experiments and shared data. Technologies like speech recognition and machine translation now rely on solid models like Hidden Markov Models (HMMs) and statistical methods. There are two aspects in HMM

  i) rigorous mathematical theory

  ii) process of training on a large corpus of real speech data.
- Neural networks, once experimental in the 1980s, are now mature and often combined with techniques from data mining and machine learning, powering a fast-growing industry.
- Judea Pearl's book in 1988 made Bayesian networks popular in AI for reasoning with uncertainty, combining ideas from classical AI and neural networks.
- This marked a shift toward probability and decision theory, supported earlier by Peter Cheeseman (1985).The idea of normative expert systems—which act rationally using decision theory—was promoted by Pearl, Horvitz, and Heckerman. Fields like robotics, computer vision, and knowledge representation have also advanced by using machine learning, reversing earlier isolation from mainstream AI.

The emergence of intelligent agents (1995- present)

- Researchers are now focusing on building complete intelligent agents, not just solving isolated AI problems. The SOAR (State,Operator,And,Result) architecture (Laird et

al., 1987; Newell, 1990) is a major example. AI is now widely used on the Internet, powering bots, search engines, and recommender systems. This shift led to:

- Better handling of uncertainty from imperfect sensors.
- Closer ties between AI and fields like control theory and economics.
- Successes in areas like robotic car control, using a mix of sensing, mapping, and planning.

- However, AI pioneers like McCarthy, Minsky, and Nilsson expressed concern that modern AI focuses too much on narrow tasks. They promote Human-Level AI (HLAI)—machines that truly think and learn—with the first symposium in 2004.
- A related area, Artificial General Intelligence (AGI), aims for systems that can learn and act in any environment. It began with ideas from Ray Solomonoff (1964) and held its first AGI conference in 2008.

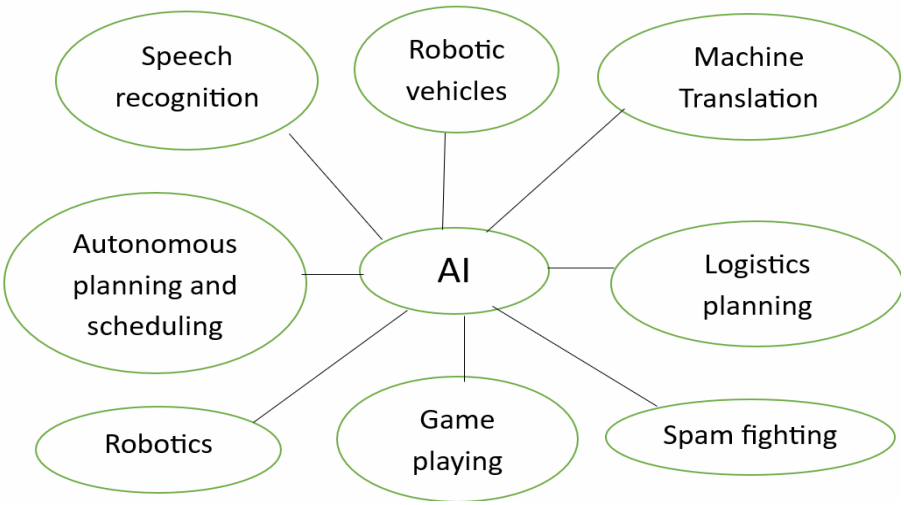
The emergence of intelligent agents (1995- present)

- However, AI pioneers like McCarthy, Minsky, and Nilsson expressed concern that modern AI focuses too much on narrow tasks. They promote Human-Level AI (HLAI)—machines that truly think and learn—with the first symposium in 2004.
- A related area, Artificial General Intelligence (AGI), aims for systems that can learn and act in any environment. It began with ideas from Ray Solomonoff (1964) and held its first AGI conference in 2008.

The availability of very large data sets (2001–present)

- 1995 – Yarowsky: Solved word meaning problems (like "plant" = tree or factory) using just definitions and large amounts of unlabeled text—no need for human-labeled examples.
- 2001 – Banko & Brill: Showed that more data improves results more than better algorithms; even simple algorithms work great with huge datasets.
- 2007 – Hays & Efros: Developed a method to fill missing parts in photos; performance jumped when dataset size grew from 10,000 to 2 million images.
- 2009 – Halevy et al.: Argued that AI's knowledge bottleneck can be solved through machine learning with lots of data, not hand-coded knowledge.
- 2005 – Kurzweil & Havenstein: Noted that thousands of AI tools were now used across industries, suggesting a new AI Spring after the past "AI Winter."

Applications of AI



Robotic vehicles:

- A driverless robotic car named STANLEY sped through the rough terrain of the Mojave dessert at 22 mph, finishing the 132-mile course first to win the 2005 DARPA Grand Challenge. STANLEY is a Volkswagen Touareg outfitted with cameras, radar, and laser rangefinders to sense the environment and onboard software to command the steering, braking, and acceleration (Thrun, 2006).
- The following year CMU's BOSS won the Urban Challenge, safely driving in traffic through the streets of a closed Air Force base, obeying traffic rules and avoiding pedestrians and other vehicles.

Speech recognition

- A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.

Autonomous planning and scheduling:

- A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft (Jonsson et al., 2000). REMOTE AGENT generated plans from high-level goals specified from the ground and monitored the execution of those plans—detecting, diagnosing, and recovering from problems as they occurred.
- Successor program MAPGEN (Al-Chang et al., 2004) plans the daily operations for NASA's Mars Exploration Rovers, and MEXAR2 (Cesta et al., 2007) did mission planning—both logistics and science planning—for the European Space Agency's Mars Express mission in 2008.

Game playing

- IBM's DEEP BLUE became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match (Goodman and Keene, 1997). Kasparov said that he felt a "new kind of intelligence" across the board from him.
- Newsweek magazine described the match as "The brain's last stand." The value of IBM's stock increased by $18 billion. Human champions studied Kasparov's loss and were able to draw a few matches in subsequent years, but the most recent human-

computer matches have been won convincingly by the computer.

Spam fighting

- Each day, learning algorithms classify over a billion messages as spam, saving the recipient from having to waste time deleting what, for many users, could comprise 80% or 90% of all messages, if not classified away by algorithms.
- Because the spammers are continually updating their tactics, it is difficult for a static programmed approach to keep up, and learning algorithms work best (Sahami et al., 1998; Goodman and Heckerman, 2004).

Logistics planning

- During the Persian Gulf crisis of 1991, U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART (Cross and Walker, 1994), to do automated logistics planning and scheduling for transportation.
- This involved up to 50,000 vehicles, cargo, and people at a time, and had to account for starting points, destinations, routes, and conflict resolution among all parameters.
- The AI planning techniques generated in hours a plan that would have taken weeks with older methods. The Defense Advanced Research Project Agency (DARPA) stated that this single application more than paid back DARPA's 30-year investment in AI.

Robotics

- The iRobot Corporation has sold over two million Roomba robotic vacuum cleaners for home use.
- The company also deploys the more rugged PackBot to Iraq and Afghanistan, where it is used to handle hazardous materials, clear explosives, and identify the location of snipers.

Machine Translation

- Computer program automatically translates from Arabic to English, allowing an English speaker to see the headline "Ardogan Confirms That Turkey Would Not Accept Any Pressure, Urging Them to Recognize Cyprus."
- The program uses a statistical model built from examples of Arabic-to-English translations and from examples of English text totaling two trillion words (Brants et al., 2007). None of the computer scientists on the team speak Arabic, but they do understand statistics and machine learning algorithms.